

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	
Pearson Edexcel Award			
Tuesday 25 June 2024			
Afternoon (Time: 3 hours)		Paper reference	9811/01
Advanced Extension Award Mathematics <i>Solutions</i>			
You must have: Mathematical Formulae and Statistical Tables An insert for questions 5, 6 and 7 (enclosed)			Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$.

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1)** or **(+S2)**.
- There are 7 questions in this question paper.
- The marks for **each** question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. In the binomial expansion of

$$(1 - 8x)^p \quad |x| < \frac{1}{8}$$

where p is a positive constant,

- the sum of the coefficient of x and the coefficient of x^2 is equal to the coefficient of x^3
- the coefficient of x^2 is positive

Determine the value of p .

(7)

Terms of expansion Coefficients required

1	
$p(-8x)$	$-8p$
$\frac{p(p-1)(-8x)^2}{2}$	$\frac{p(p-1) \times 64}{2}$
$\frac{p(p-1)(p-2)(-8x)^3}{6}$	$\frac{p(p-1)(p-2) \times -512}{6}$

$$-8p + 32p(p-1) = \frac{-512p(p-1)(p-2)}{6} \quad (\div 8)$$

$$-p + 4p(p-1) = \frac{-64p(p-1)(p-2)}{6} \quad (\times 6)$$

$$-6p + 24p(p-1) = -64p(p-1)(p-2) \quad (\div 2p)$$

$$-3 + 12(p-1) = -32(p-1)(p-2)$$

$$12p - 15 = -32(p^2 - 3p + 2)$$

$$12p - 15 = -32p^2 + 96p - 64$$

$$32p^2 - 84p + 49 = 0$$

$$(ax - 7)(bx - 7) = 0$$

Since the coefficient of p is negative the factors will have a constant term of -7
Since $-7 \times -7 = +49$



Question 1 continued

$$ab = 32 \quad -7a - 7b = -84$$

Try 1 and 32 $-7 - 7 \times 32 \neq -84$

2 and 16 $-14 - 7 \times 16 = -14 - 112 \neq -84$

4 and 8 $-28 - 56 = -84$ ✓

$$32p^2 - 84p + 49 = 0$$

$$(4x-7)(8x-7) = 0$$

$$x = \frac{7}{4} \quad x = \frac{7}{8}$$

Since coefficient of $x^2 > 0$ $32p(p-1) > 0$
 $p(p-1) > 0$

If $x = \frac{7}{4}$ $\frac{7}{4} \left(\frac{3}{4} \right) = \frac{21}{16} > 0$

If $x = \frac{7}{8}$ $\frac{7}{8} \left(-\frac{1}{8} \right) = -\frac{7}{64} < 0$

So $x = \frac{7}{4}$

(Total for Question 1 is 7 marks)

2.

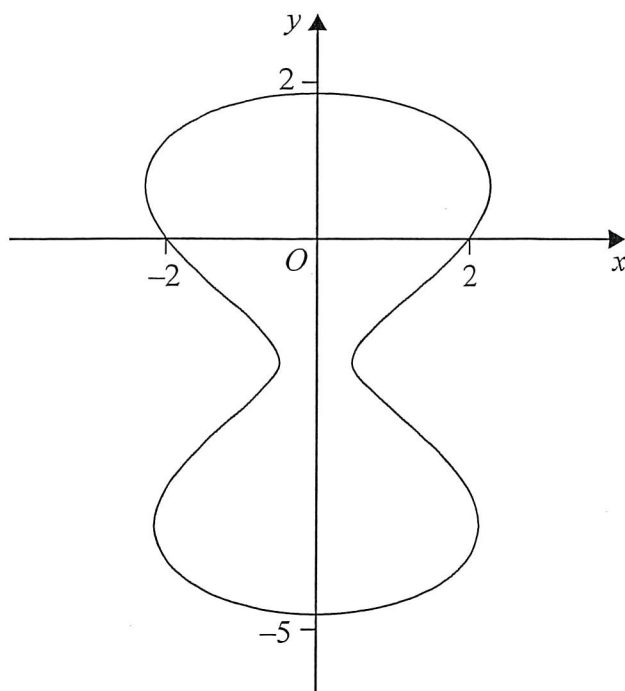


Figure 1

Figure 1 shows the curve defined by the equation

$$y^2 + 3y - 6\sin y = 4 - x^2$$

The point $P(x, y)$ lies on the curve.

The distance from the origin, O , to P is D .

(a) Write down an equation for D^2 in terms of y only.

(1)

(b) Hence determine the minimum value of D giving your answer in simplest form.

(5)

$$a) \quad D^2 = x^2 + y^2$$

↓

$$y^2 + 3y - 6\sin y = 4 - x^2 \Rightarrow x^2 = 4 - y^2 - 3y + 6\sin y$$

$$D^2 = 4 - y^2 - 3y + 6\sin y + y^2$$

$$D^2 = 4 - 3y + 6\sin y$$



Question 2 continued

b) At the minimum $\frac{dD}{dy} = 0$ so $\frac{dD^2}{dy} = 0$

$$D^2 = 4 - 3y + 6\sin y$$

$$\frac{dD^2}{dy} = -3 + 6\cos y = 0$$

$$-5 < y < 2$$

$$6\cos y = 3$$

$$\cos y = 1/2$$

$$y = \cos^{-1}(1/2) = \frac{\pi}{3} \text{ or } -\frac{\pi}{3}$$

From the graph the minimum occurs when $y < 0$

$$\Rightarrow y = -\pi/3$$

$$D^2 = 4 - 3\left(-\frac{\pi}{3}\right) + 6\sin\left(-\frac{\pi}{3}\right)$$

$$D^2 = 4 + \pi + 6 \times \frac{-\sqrt{3}}{2}$$

$$D^2 = 4 + \pi - 3\sqrt{3}$$

$$D = \sqrt{4 + \pi - 3\sqrt{3}}$$

(Total for Question 2 is 6 marks)

3. (i) Determine the value of k such that

$$\arctan \frac{1}{2} - \arctan \frac{1}{3} = \arctan k \quad (3)$$

(ii)(a) Prove that

$$\cos 3A \equiv 4\cos^3 A - 3\cos A \quad (3)$$

Given that $a = \cos 20^\circ$

(b) write down, in terms of a , an expression for $\cos 40^\circ$ (1)

(c) determine, in terms of a , a simplified expression for $\cos 80^\circ$ (2)

(d) Use part (a) to show that

$$4a^3 - 3a = \frac{1}{2} \quad (1)$$

(e) Hence, or otherwise, show that

$$\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8} \quad (4)$$

$$i) \arctan(k) = \arctan(1/2) - \arctan(1/3)$$

$$k = \tan(\arctan(1/2) - \arctan(1/3))$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$A = \arctan 1/2$$

$$B = \arctan 1/3$$

$$k = \frac{\tan \arctan(1/2) - \tan \arctan(1/3)}{1 + \tan \arctan(1/2) \tan \arctan(1/3)}$$

$$k = \frac{1/2 - 1/3}{1 + 1/2 \times 1/3} = \frac{1/6}{1 + 1/6} = \frac{1}{6} \div \frac{7}{6} = \frac{1}{6} \times \frac{6}{7} = \frac{1}{7}$$



Question 3 continued

$$\text{ii a) } \cos(3A) = \cos(2A + A) = \cos(2A)\cos(A) - \sin(2A)\sin(A)$$

$$\cos(2A) = 2\cos^2(A) - 1 \quad \Rightarrow (2\cos^2(A) - 1)\cos(A) - 2\sin(A)\cos(A)\sin(A)$$

$$\sin(2A) = 2\sin(A)\cos(A) \quad \Rightarrow 2\cos^3(A) - \cos(A) - 2\sin^2(A)\cos(A)$$

$$= 2\cos^3(A) - \cos(A) - 2(1 - \cos^2(A))\cos(A)$$

$$= 2\cos^3(A) - \cos(A) - 2\cos(A) + 2\cos^3(A)$$

$$= 4\cos^3(A) - 3\cos(A)$$

$$\text{ii b) } \cos(40) = \cos(2 \times 20) = 2\cos^2(20) - 1$$

$$= 2a^2 - 1$$

$$\text{ii c) } \cos(80) = \cos(2 \times 40) = 2\cos^2(40) - 1$$

$$= 2(2a^2 - 1)^2 - 1$$

$$= 2(4a^4 - 4a^2 + 1) - 1$$

$$= 8a^4 - 8a^2 + 1$$

$$\text{ii d) } a = \cos(20) \quad \text{from (a)}$$

$$4a^3 - 3a = 4\cos^3(20) - 3\cos(20) = \cos(3 \times 20) = \cos(60) = \frac{1}{2}$$

$$\text{ii e) } \cos(20)\cos(40)\cos(80) = a(2a^2 - 1)(8a^4 - 8a^2 + 1)$$

$$= (2a^3 - a)(8a^4 - 8a^2 + 1)$$

Need to use (a) so rewrite in terms of $4a^3 - 3a$

$$2a^3 - a \rightarrow \frac{1}{2}(4a^3 - 3a) + \frac{1}{2}a = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2}a = \frac{1}{4} + \frac{1}{2}a$$

$$8a^4 - 8a^2 + 1 \rightarrow 2a(4a^3 - 3a) - 2a^2 + 1 = 2a \times \frac{1}{2} - 2a^2 + 1$$

$$= a - 2a^2 + 1$$

$$(2a^3 - a)(8a^4 - 8a^2 + 1) = \left(\frac{1}{4} + \frac{1}{2}a\right)(a - 2a^2 + 1)$$

$$= \frac{1}{4}(1 + 2a)(a - 2a^2 + 1) = \frac{1}{4}(a - 2a^2 + 1 + 2a^2 - 4a^3 + 2a) = \frac{1}{4}(-4a^3 + 3a + 1)$$

$$= \frac{1}{4}(1 - (4a^3 - 3a)) = \frac{1}{4}(1 - \frac{1}{2}) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$$



Question 3 continued

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Question 3 continued

(Total for Question 3 is 14 marks)

4. (a) Use the substitution $x = \sqrt{3} \tan u$ to show that

$$\int \frac{1}{3+x^2} dx = p \arctan(px) + c$$

where p is a real constant to be determined and c is an arbitrary constant.

(5)

- (b) Use the substitution $x = \frac{3u+3}{u-3}$ to determine the exact value of I where

$$I = \int_{-3}^1 \frac{\ln(3-x)}{3+x^2} dx$$

giving your answer in simplest form.

(10)

(+S1)

$$c) \quad x = \sqrt{3} \tan(u) \quad \frac{dx}{du} = \sqrt{3} \sec^2(u) \quad x^2 = 3 \tan^2(u)$$

$$\int \frac{1}{3+x^2} dx = \int \frac{1}{3+3\tan^2(u)} \sqrt{3} \sec^2(u) du$$

$$= \int \frac{\sqrt{3} \sec^2(u)}{3 \sec^2(u)} du$$

$$1 + \tan^2(u) = \sec^2(u)$$

$$3 + 3 \tan^2(u) = 3 \sec^2(u)$$

$$= \int \frac{\sqrt{3}}{3} du = \frac{\sqrt{3}}{3} u + c = \frac{\sqrt{3}}{3} \arctan\left(\frac{x}{\sqrt{3}}\right) + c$$

$$x = \sqrt{3} \tan(u)$$

$$= \frac{\sqrt{3}}{3} \arctan\left(\frac{\sqrt{3} x}{3}\right) + c$$

$$\frac{x}{\sqrt{3}} = \tan(u)$$

$$\arctan\left(\frac{x}{\sqrt{3}}\right) = u$$



Question 4 continued

$$\begin{aligned}
 \text{b) } x &= \frac{3u+3}{u-3} & \frac{dx}{du} &= \frac{(u-3)(3u+3)' - (3u+3)(u-3)'}{(u-3)^2} \\
 & & &= \frac{3(u-3) - 1(3u+3)}{(u-3)^2} \\
 & & &= \frac{3u-9-3u-3}{(u-3)^2} = \frac{-12}{(u-3)^2}
 \end{aligned}$$

$$\begin{aligned}
 3-x &= 3 - \frac{3u+3}{u-3} = \frac{3(u-3) - (3u+3)}{u-3} = \frac{3u-9-3u-3}{u-3} \\
 & & &= \frac{-12}{u-3} = \frac{12}{3-u}
 \end{aligned}$$

$$x^2 = \left(\frac{3u+3}{u-3} \right)^2 = \frac{9u^2+18u+9}{(u-3)^2}$$

$$\begin{aligned}
 3+x^2 &= 3 + \frac{9u^2+18u+9}{(u-3)^2} = \frac{3(u-3)^2 + 9u^2+18u+9}{(u-3)^2} \\
 &= \frac{3(u^2-6u+9)+9u^2+18u+9}{(u-3)^2} = \frac{12u^2+36}{(u-3)^2}
 \end{aligned}$$

$$\begin{aligned}
 \text{When } x=1 & \quad 1 = \frac{3u+3}{u-3} \Rightarrow u-3=3u+3 \\
 & \quad \quad \quad -6=2u \Rightarrow u=-3
 \end{aligned}$$

$$\begin{aligned}
 \text{When } x=-3 & \quad -3 = \frac{3u+3}{u-3} \Rightarrow -3(u-3)=3u+3 \\
 & \quad \quad \quad -3u+9=3u+3 \\
 & \quad \quad \quad 6=6u \Rightarrow u=1
 \end{aligned}$$

$$\begin{aligned}
 \int_{-3}^1 \frac{\ln(3-x)}{3+x^2} dx &= \int_1^{-3} \frac{\ln\left(\frac{12}{3-u}\right)}{\frac{12u^2+36}{(u-3)^2}} \cdot \frac{-12}{(u-3)^2} du \\
 &= \int_1^{-3} \frac{\ln\left(\frac{12}{3-u}\right)}{12u^2+36} (u-3)^2 \cdot \frac{-12}{(u-3)^2} du
 \end{aligned}$$

Question 4 continued

$$\int_a^b f(x) = - \int_b^a f(x)$$

$$\int_1^{-3} \frac{-12 \ln\left(\frac{12}{3-u}\right)}{12(u^2+3)} du = \int_{-3}^1 \frac{\ln\left(\frac{12}{3-u}\right)}{u^2+3} du$$

$$= \int_{-3}^1 \frac{\ln(12)}{u^2+3} du - \int_{-3}^1 \frac{\ln(3-u)}{u^2+3} du$$

Let original integral = I $\rightarrow I$

$$I = \int_{-3}^1 \frac{\ln(12)}{u^2+3} du - I$$

$$2I = \left[\frac{\sqrt{3}}{3} \arctan\left(\frac{\sqrt{3}u}{3}\right) \right]_{-3}^1 \cdot \ln(12)$$

$$2I = \ln(12) \left(\frac{\sqrt{3}}{3} \arctan\left(\frac{\sqrt{3}}{3}\right) - \frac{\sqrt{3}}{3} \arctan\left(-\sqrt{3}\right) \right)$$

$$2I = \frac{\sqrt{3}}{3} \ln(12) \left(\frac{\pi}{6} + \frac{\pi}{3} \right) = \frac{\pi \sqrt{3} \ln(12)}{6}$$

$\div 2$

$$I = \frac{\pi \sqrt{3} \ln(12)}{12}$$

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Question 4 continued

(Total for Question 4 is 16 marks)

5.

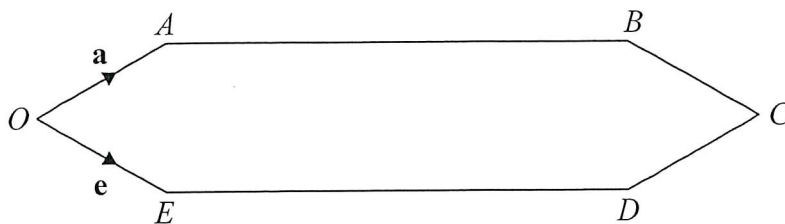


Figure 2

Figure 2 shows a sketch of a hexagon $OABCDE$ where

- the interior angle at O and at C are each 60°
- the interior angle at each of the other vertices is 150°
- $OA = OE = BC = CD$
- $AB = ED = 3 \times OA$

Given that $\vec{OA} = \mathbf{a}$ and $\vec{OE} = \mathbf{e}$

determine as simplified expressions in terms of $\mathbf{a}$ and $\mathbf{e}$

(a) $\vec{AB}$ (3)

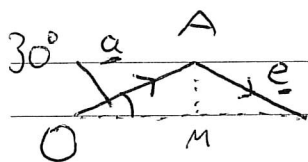
(b) $\vec{OD}$ (2)

The point R divides AB internally in the ratio $1:2$

(c) Determine $\vec{RC}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{e}$ (2)

The line through the points R and C meets the line through the points O and D at the point X .

(d) Determine $\vec{OX}$ in the form $\lambda\mathbf{a} + \mu\mathbf{e}$, where λ and μ are real values in simplest form. (6)
(+S2)



$$\vec{AB} = \lambda(\underline{a} + \underline{e})$$

$$|OM| = |\underline{a}| \cos(30^\circ) = \frac{\sqrt{3}}{2} |\underline{a}|$$

$$|\underline{a} + \underline{e}| = \frac{\sqrt{3}}{2} |\underline{a}| \times 2 = \sqrt{3} |\underline{a}|$$

$$\text{Since } |AB| = 3|\underline{a}| \quad |AB| = \sqrt{3}(\sqrt{3}|\underline{a}|) = \sqrt{3} |\underline{a} + \underline{e}|$$

$$\Rightarrow \vec{AB} = \sqrt{3}(\underline{a} + \underline{e})$$



Question 5 continued

$$b) \vec{OD} = \vec{OE} + \vec{ED} = \vec{OE} + \vec{AB} = \underline{e} + \sqrt{3}(\underline{a} + \underline{e}) \\ = \sqrt{3}\underline{a} + (1 + \sqrt{3})\underline{e}$$

$$c) AR : RB \Rightarrow RB = \frac{2}{3} AB$$

$$\vec{RC} = \vec{RB} + \vec{BC} = \frac{2}{3} \vec{AB} + \vec{BC} = \frac{2}{3} \vec{AB} + \vec{OE}$$

$$= \frac{2}{3} \sqrt{3}(\underline{a} + \underline{e}) + \underline{e} = \frac{2\sqrt{3}}{3} \underline{a} + \left(\frac{2\sqrt{3}}{3} + 1\right) \underline{e}$$

$$d) \vec{OX} = \vec{OA} + \vec{AR} + \mu \vec{RC}$$

$$= \underline{a} + \frac{\sqrt{3}}{3}(\underline{a} + \underline{e}) + \mu \frac{2\sqrt{3}}{3} \underline{a} + \mu \left(\frac{2\sqrt{3}}{3} + 1\right) \underline{e}$$

$$\vec{OX} = \gamma \vec{OD} = \gamma \sqrt{3} \underline{a} + \gamma(1 + \sqrt{3}) \underline{e}$$

Equate $\underline{a}$ $1 + \frac{\sqrt{3}}{3} + \mu \frac{2\sqrt{3}}{3} = \gamma \sqrt{3}$

$$(\div \sqrt{3})$$

$$\frac{\sqrt{3}}{3} + \frac{1}{3} + \frac{2}{3} \mu = \gamma$$

Equate $\underline{e}$ $\frac{\sqrt{3}}{3} + \mu \left(\frac{2\sqrt{3}}{3} + 1\right) = \gamma(1 + \sqrt{3})$

$$\frac{\sqrt{3}}{3} + \mu \left(\frac{2\sqrt{3}}{3} + 1\right) = \left(\frac{\sqrt{3}}{3} + \frac{1}{3} + \frac{2}{3} \mu\right)(1 + \sqrt{3})$$

$$(\times 3)$$

$$\sqrt{3} + \mu(2\sqrt{3} + 3) = (\sqrt{3} + 1 + 2\mu)(1 + \sqrt{3})$$

$$\sqrt{3} + \mu(2\sqrt{3} + 3) = \sqrt{3} + 1 + 2\mu + 3 + \sqrt{3} + 2\sqrt{3}\mu$$

$$\mu(2\sqrt{3} + 3 - 2 - 2\sqrt{3}) = \sqrt{3} + 1 + 3 + \sqrt{3} - \sqrt{3} \\ = 4 + \sqrt{3}$$

$$\vec{OX} = \underline{a} + \frac{\sqrt{3}}{3}(\underline{a} + \underline{e}) + (4 + \sqrt{3}) \frac{2\sqrt{3}}{3} \underline{a} + (4 + \sqrt{3}) \left(\frac{2\sqrt{3}}{3} + 1\right) \underline{e}$$

$$= \underline{a} + \frac{\sqrt{3}}{3} \underline{a} + \frac{\sqrt{3}}{3} \underline{e} + \frac{8\sqrt{3}}{3} \underline{a} + 2\underline{a} + \frac{8\sqrt{3}}{3} \underline{e} + 4\underline{e} + 2\underline{e} + \sqrt{3}\underline{e}$$

$$= (3 + 3\sqrt{3}) \underline{a} + (6 + 4\sqrt{3}) \underline{e}$$



Question 5 continued

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Question 5 continued

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Question 5 continued

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Lined area for writing the answer to Question 5.



Question 5 continued

(Total for Question 5 is 15 marks)

6.

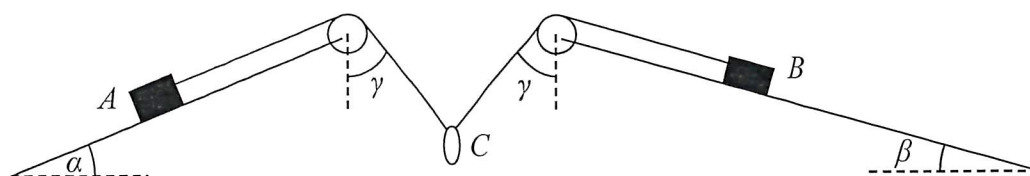


Figure 3

Figure 3 shows a block A with mass $4m$ and a block B with mass $5m$.
Block A is at rest on a rough plane inclined at an angle α to the horizontal.
Block B is at rest on a rough plane inclined at an angle β to the horizontal.

The blocks are connected by a light inextensible string which passes over a small smooth pulley at the top of each plane.

A small smooth ring C , of mass $8m$, is threaded on the string between the pulleys so that A , B and C all lie in the same vertical plane.

The part of the string between A and its pulley lies along a line of greatest slope of the plane of angle α .

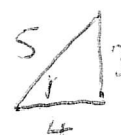
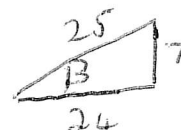
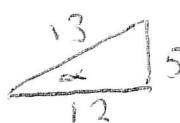
The part of the string between B and its pulley lies along a line of greatest slope of the plane of angle β .

The angle between the vertical and the string between each pulley and the ring C is γ .

The two blocks, A and B , are modelled as particles.

Given that

- $\tan \alpha = \frac{5}{12}$ and $\tan \beta = \frac{7}{24}$ and $\tan \gamma = \frac{3}{4}$
- the coefficient of friction, μ , is the same between each block and its plane
- one of the blocks is on the point of sliding up its plane
- the tension in the string is T



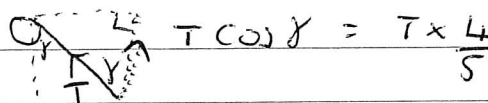
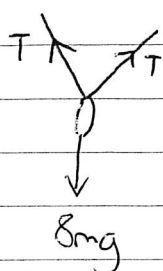
(a) determine, in terms of m and g , an expression for T ,

(3)

(b) draw a diagram showing the forces on block A , clearly labelling each of the forces acting on the block,

(3)

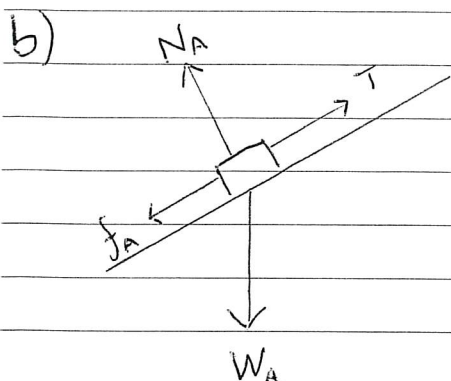
(c) determine the value of μ , giving a justification for your answer.

(10)
(+S2)

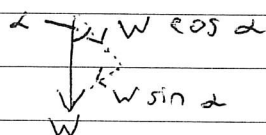
$$\frac{4}{5}T + \frac{4}{5}T = 8mg \Rightarrow \frac{8T}{5} = 8mg \Rightarrow T = 5mg$$



Question 6 continued



N_A is the normal reaction
 T is the tension in the string
 W_A is the weight of block A
 f_A is the friction between the slope and block A.



c) Block A: parallel to plane

$$T - f_A - 4mg \sin \alpha = m \times 0$$

$$5mg - 4mg \left(\frac{5}{13}\right) = f_A$$

$$\frac{45}{13} mg = f_A$$

perpendicular to plane

$$N_A = 4mg \cos \alpha = 4mg \left(\frac{12}{13}\right)$$

$$N_A = \frac{48}{13} mg$$

One block is on the point of sliding so for one block
 $f = \mu R$

$$\mu = \frac{f}{R} = \frac{\frac{45}{13} mg}{\frac{48}{13} mg} = \frac{45}{48} = \frac{15}{16}$$

for block B the equations would be of the same form but with
 $W = 5mg$ and angle β $\therefore f_B = 5mg - 5mg \left(\frac{7}{25}\right) = \frac{18}{5} mg$

$$N_B = 5mg \left(\frac{24}{25}\right) = \frac{24}{5} mg$$

$$\mu_B = \frac{\frac{18}{5} mg}{\frac{24}{5} mg} = \frac{18}{24} = \frac{3}{4}$$

Since $\frac{3}{4} < \frac{15}{16}$ $\mu = \frac{15}{16}$

If $\mu = \frac{3}{4}$ $\mu N_A = \frac{36}{13} mg$ so $f_A < \mu N_A$ hence not at

maximum and thus not on the point of sliding, the system
 would not move.

Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

Question 6 continued

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Question 6 continued

(Total for Question 6 is 18 marks)

7.

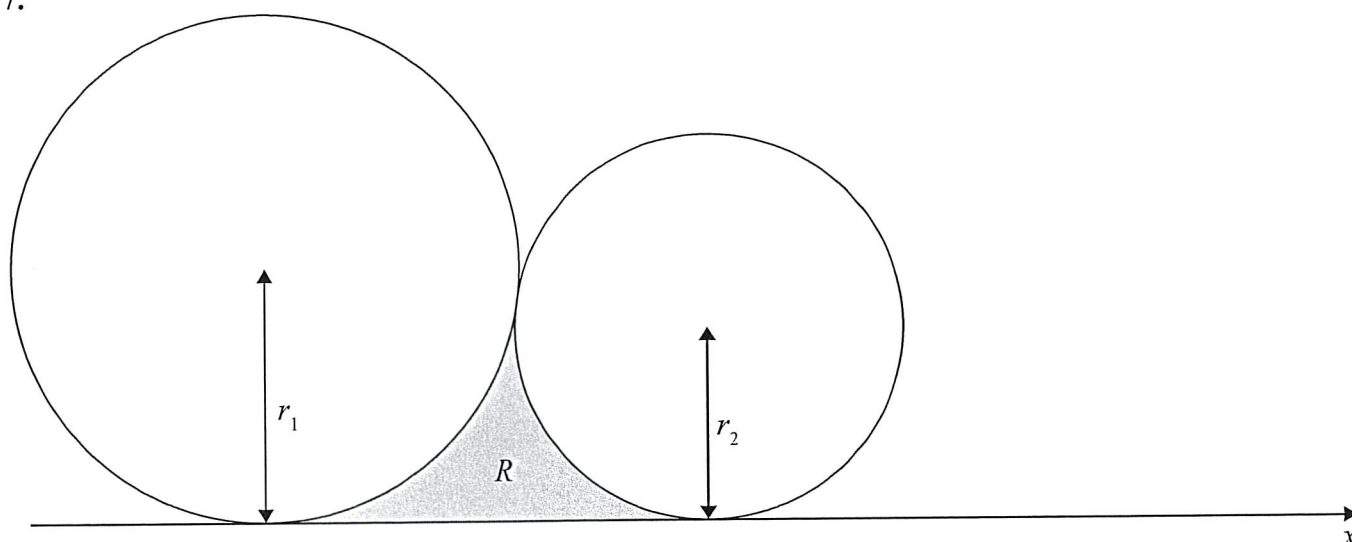


Figure 4

Figure 4 shows a circle with radius r_1 and a circle with radius r_2 . The circles touch externally at a single point above the x -axis. Both circles also have the x -axis as a tangent.

- (a) Show that the horizontal distance between the centres of the circles, d , is given by

$$d^2 = 4r_1 r_2 \quad (2)$$

The finite region R , shown shaded in Figure 4, is bounded by the x -axis and minor arcs of the two circles.

Given that $r_1 \geq r_2$

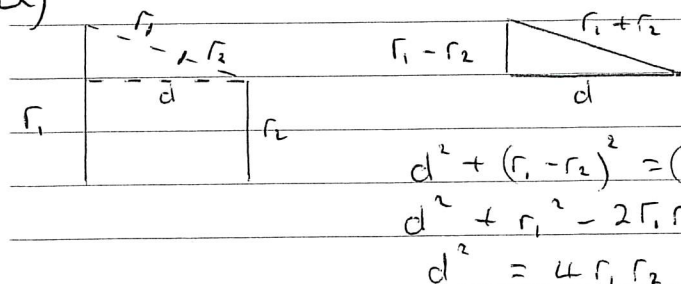
- (b) show that the area of R is given by

$$(r_1 + r_2)\sqrt{r_1 r_2} - \frac{1}{2}(r_1^2 - r_2^2)\theta - \frac{1}{2}\pi r_2^2$$

where $\cos \theta = \frac{r_1 - r_2}{r_1 + r_2}$ (5)

Question 7 continues on the next page.

a)



Question 7 continued

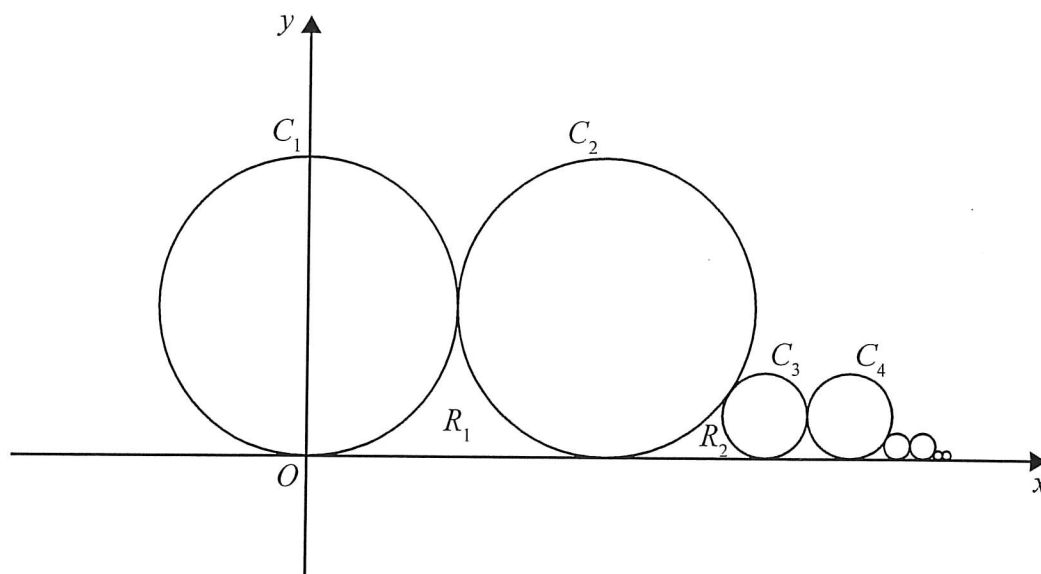


Figure 5

A sequence of circles, $C_1, C_2, C_3, \dots$ with radii $r_1, r_2, r_3, \dots$ respectively, is constructed such that

- each circle is tangential to and above the x -axis
- the first circle, C_1 , has centre $(0, 1)$
- each successive circle touches the preceding one externally at a single point
- the horizontal distances between the centres of successive circles form a geometric sequence with first term 2 and common ratio $\frac{1}{\sqrt{3}}$

The first few circles in the sequence are shown in Figure 5.

(c) (i) Determine the value of r_3

(ii) Show that, for $n \geq 1$, $r_{n+2} = kr_n$ where k is a constant to be determined.

(iii) Hence show that, for $n \geq 1$, $r_{2n} = r_{2n-1}$

(9)

The region bounded between C_n, C_{n+1} and the x -axis is R_n

The total area, A , bounded above the x -axis and under all the circles is the sum of the areas of all these regions.

(d) Determine the value of A , giving the answer in simplest form.

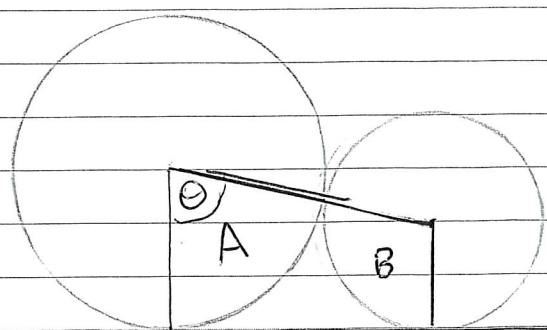
(6)

(+S2)



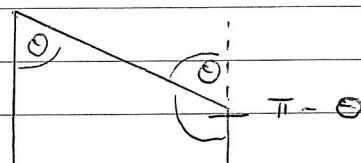
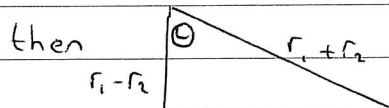
Question 7 continued

b)



$$R = \text{trapezium} - \text{sector A} - \text{sector B}$$

$$\text{If } \cos \theta = \frac{r_1 - r_2}{r_1 + r_2} = \frac{\text{adj}}{\text{hyp}}$$



$$\text{trapezium} = \frac{r_1 + r_2}{2} \times d = \frac{r_1 + r_2}{2} \times \sqrt{4r_1 r_2} = (r_1 + r_2) \sqrt{r_1 r_2}$$

$$\text{Sector A} = \frac{1}{2} r_1^2 \theta$$

$$\text{Sector B} = \frac{1}{2} r_2^2 (\pi - \theta)$$

$$R = (r_1 + r_2) \sqrt{r_1 r_2} - \frac{1}{2} r_1^2 \theta - \left(\frac{1}{2} r_2^2 \pi - \frac{1}{2} r_2^2 \theta \right)$$

$$= (r_1 + r_2) \sqrt{r_1 r_2} - \frac{1}{2} r_1^2 \theta + \frac{1}{2} r_2^2 \theta - \frac{1}{2} r_2^2 \pi$$

$$= (r_1 + r_2) \sqrt{r_1 r_2} - \frac{1}{2} (r_1^2 - r_2^2) \theta - \frac{1}{2} \pi r_2^2$$

c i) distance between C_2 and $C_1 = 2$ Since $r_1 = 1$, $r_2 = 1$

distance between C_3 and $C_2 = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$

from (a) $d^2 = 4r_1 r_2$

$$\left(\frac{2\sqrt{3}}{3} \right)^2 = 4r_2 r_3 \Rightarrow \frac{4}{3} = 4 \times 1 \times r_3 \Rightarrow \frac{1}{3} = r_3$$



Question 7 continued

Cii) Let distance between C_n and C_{n+1} be d_n
then distance between C_{n+1} and C_{n+2} is d_{n+1}

$$d_{n+1} = \frac{1}{\sqrt{3}} d_n$$

$$(d_{n+1})^2 = \left(\frac{1}{\sqrt{3}} d_n\right)^2 = \frac{1}{3} d_n^2$$

$$4r_{n+1} r_{n+2} = \frac{1}{3} \cdot 4r_n r_{n+1}$$

$$4r_{n+1}$$

$$r_{n+2} = \frac{1}{3} r_n$$

Ciii) $r_1 = 1$ $r_3 = \frac{1}{3}$ $r_5 = \frac{1}{9}$ $r_7 = \frac{1}{27}$... from (cii)

Likewise $r_2 = 1$ $r_4 = \frac{1}{3}$ $r_6 = \frac{1}{9}$...

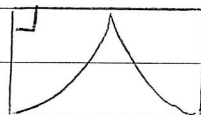
Each form a geometric sequence with $a=1$ $r=\frac{1}{3}$

$$r_{\text{odd}} = r_{2n-1} = 1 \times \left(\frac{1}{3}\right)^{n-1} = \left(\frac{1}{3}\right)^{n-1}$$

$$r_{\text{even}} = r_{2n} = 1 \times \left(\frac{1}{3}\right)^{n-1} = \left(\frac{1}{3}\right)^{n-1} \quad \therefore r_{2n} = r_{2n-1} \quad n \geq 1$$

d) For areas R_1 R_3 R_5 etc the adjacent circles have equal radii

$$R_{2n-1} \quad r = \left(\frac{1}{3}\right)^{n-1}$$



$$\odot = \frac{\pi}{2}$$

from (b)

$$\begin{aligned} R_{2n-1} &= \left(\frac{1}{3}^{n-1} \times 2\right) \sqrt{\left(\frac{1}{3}^{n-1}\right)^2} - 0 - \frac{1}{2} \pi \left(\frac{1}{3}^{n-1}\right)^2 \\ &= \left(\frac{1}{3}\right)^{n-1} \left(2 \left(\frac{1}{3}\right)^{n-1} - \frac{1}{2} \pi \left(\frac{1}{3}\right)^{n-1}\right) \\ &= \left(\frac{1}{3}\right)^{n-1} \left(2 - \frac{\pi}{2}\right) \\ &= \left(\frac{1}{9}\right)^{n-1} \left(2 - \frac{\pi}{2}\right) \end{aligned}$$

Question 7 continued

for areas R_2, R_4, R_6 etc. the adjacent circles have different radii.

for R_{2n} radius of larger circle = $\left(\frac{1}{3}\right)^{n-1}$

radius of smaller circle = $\frac{1}{3} \times \left(\frac{1}{3}\right)^{n-1}$

from (b) $\cos \theta = \frac{r_1 - r_2}{r_1 + r_2}$

$$\cos \theta = \frac{\left(\frac{1}{3}\right)^{n-1} - \frac{1}{3} \times \left(\frac{1}{3}\right)^{n-1}}{\left(\frac{1}{3}\right)^{n-1} + \frac{1}{3} \times \left(\frac{1}{3}\right)^{n-1}} = \frac{\frac{2}{3} \left(\frac{1}{3}\right)^{n-1}}{\frac{4}{3} \left(\frac{1}{3}\right)^{n-1}} = \frac{2}{3} \div \frac{4}{3} = \frac{2}{4} = \frac{1}{2}$$

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) \quad \theta = \frac{\pi}{3}$$

from (b)

$$\text{Area } R_{2n} = \left(\left(\frac{1}{3}\right)^{n-1} + \frac{1}{3} \times \left(\frac{1}{3}\right)^{n-1}\right) \sqrt{\frac{1}{3} \times \left(\frac{1}{3}\right)^{n-1}}^2 - \frac{1}{2} \left(\left(\frac{1}{3}\right)^{n-1} - \frac{1}{3} \times \left(\frac{1}{3}\right)^{n-1}\right)^2 \frac{\pi}{3} - \frac{\pi}{2} \left(\frac{1}{3} \times \left(\frac{1}{3}\right)^{n-1}\right)^2$$

$$= \frac{4}{3} \times \left(\frac{1}{3}\right)^{n-1} \times \sqrt{\frac{1}{3}} \times \left(\frac{1}{3}\right)^{n-1} - \frac{\pi}{6} \left(\left(\frac{1}{9}\right)^{n-1} - \frac{1}{9} \times \left(\frac{1}{9}\right)^{n-1}\right) - \frac{\pi}{2} \times \frac{1}{9} \times \left(\frac{1}{9}\right)^{n-1}$$

$$= \frac{4\sqrt{3}}{9} \times \left(\frac{1}{9}\right)^{n-1} - \frac{\pi}{6} \times \frac{8}{9} \times \left(\frac{1}{9}\right)^{n-1} - \frac{\pi}{18} \times \left(\frac{1}{9}\right)^{n-1}$$

$$= \left(\frac{1}{9}\right)^{n-1} \left(\frac{4\sqrt{3}}{9} - \frac{8\pi}{54} - \frac{\pi}{18}\right) = \left(\frac{1}{9}\right)^{n-1} \left(\frac{4\sqrt{3}}{9} - \frac{11\pi}{54}\right)$$

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Question 7 continued

For R_1 R_{2n-1} $n=1$ Area = $2 - \pi/2$

R_3 R_{2n-1} $n=2$ Area = $\frac{1}{9} (2 - \pi/2)$

R_5 R_{2n-1} $n=3$ Area = $\frac{1}{9^2} (2 - \pi/2)$

Geometric series $a = 2 - \pi/2$ $r = 1/9$

$$S_{\infty} = \frac{2 - \pi/2}{1 - 1/9} = (2 - \frac{\pi}{2}) \div \frac{8}{9} = (2 - \frac{\pi}{2}) \times \frac{9}{8}$$

For R_2 R_{2n} $n=1$ Area = $\frac{4\sqrt{3}}{9} - \frac{11\pi}{54}$

R_4 R_{2n} $n=2$ Area = $\frac{1}{9} \left(\frac{4\sqrt{3}}{9} - \frac{11\pi}{54} \right)$

Again geometric series $a = \frac{4\sqrt{3}}{9} - \frac{11\pi}{54}$ $r = \frac{1}{9}$

$$S_{\infty} = \frac{\frac{4\sqrt{3}}{9} - \frac{11\pi}{54}}{1 - 1/9} = \left(\frac{4\sqrt{3}}{9} - \frac{11\pi}{54} \right) \div \frac{8}{9} = \left(\frac{4\sqrt{3}}{9} - \frac{11\pi}{54} \right) \times \frac{9}{8}$$

$$\text{Total area} = \frac{9}{8} \left(2 - \frac{\pi}{2} + \frac{4\sqrt{3}}{9} - \frac{11\pi}{54} \right) = \frac{9}{8} \left(2 + \frac{4\sqrt{3}}{9} - \frac{38\pi}{54} \right)$$

$$= \frac{9}{4} + \frac{\sqrt{3}}{2} - \frac{19\pi}{24}$$

Question 7 continued

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(Total for Question 7 is 24 marks)

TOTAL FOR PAPER IS 100 MARKS

